Overview of the Simulation Conditions

The statistical tests evaluate the relative performance of 5 maximum likelihood estimation methods (labeled **(i)** through **(v)**) for estimating population proportions from simulated composition data based on their Root Mean Square Error (RMSE). The 5 methods are:

(i) Dirichlet-Multinomial (DM) likelihood with unconstrained concentration vector $\underline{\alpha}$

(ii) DM likelihood with constrained total concentration α_0

(iii) DM likelihood with fixed linear scaling parameter θ

(iv) Multinomial likelihood

(v) DM likelihood freely-estimated linear scaling parameter θ

These methods were tested under a specific error and sampling regime, which included a target mean sample size of 100 drawn from a Negative binomial distribution, an overdispersion multiplier of 0.5, a base true theta of 10 with a CV of 0.2, and a log-scale observation error (sigma) of 0.5. Because the error parameters are fixed in this specific run, the statistical tests evaluate *how well each method handles this specific combination of observation error and overdispersion*, rather than testing the parameters themselves.

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1. Friedman Rank Sum Test

The Friedman test evaluates whether the five estimation methods rank consistently differently in their RMSE across all simulation runs.

- **Hypothesis Testing:** The null hypothesis assumes that all five estimation methods perform equally well, meaning their median RMSE values are not significantly different. The alternative hypothesis is that at least one method consistently ranks differently than the others.
- **Results:** The test yielded a chi-squared value of 290.64 with a p-value of < 0.0001.
- **Conclusion:** Because the p-value is practically zero, we reject the null hypothesis and conclude that the choice of estimation method significantly impacts model performance. However, Kendall's W is approximately 0.031, indicating that while the differences are statistically significant, the overall effect size or agreement in rankings across the simulation runs is relatively small.

2. Pairwise Wilcoxon Signed-Rank Tests

The Wilcoxon tests break down the omnibus Friedman test by comparing the estimation methods head-to-head to determine exactly which methods outperform the others.

- **Hypothesis Testing:** For each pair of methods (e.g., Method **(i)** vs. Method **(ii)**), the null hypothesis states that the true location shift (the median difference in their RMSEs) is zero. The alternative hypothesis is that the true location shift is not equal to zero.
- **Results: * Method (ii) is the worst performer:** Method **(ii)** had a significantly higher RMSE than all other methods ($p < 0.0001$ for comparisons against **(i)**, **(iii)**, **(iv)**, and **(v)**).
  - **Method (v) vs. (iii) and (iv):** Method **(v)** showed slight but statistically significant improvements over Method **(iii)** ($p = 0.003258$) and Method **(iv)** ($p = 2.902e-05$).
  - **Indistinguishable pairs:** There was no statistically significant difference between Method **(iii)** and **(i)** ($p = 0.8478$), **(iv)** and **(i)** ($p = 0.5339$), or **(iv)** and **(iii)** ($p = 0.8435$).

3. Linear Mixed Effects Model (LMM) and Estimated Marginal Means (emmeans)

The LMM approach models the RMSE while accounting for the random effect of the specific simulation run, providing robust estimates of average performance.

- **Hypothesis Testing:** The null hypothesis for the ANOVA on the mixed model is that the fixed effect of the method does not explain any variance in the RMSE. The contrasts test whether the difference in estimated marginal means between any two methods is zero.
- **Results:** The Type III ANOVA confirms a highly significant effect of the method on RMSE ($F = 149.75$, $p < 0.0001$). The estimated marginal means (emmeans) for the RMSE of each method are:
  - Method **(i)**: 0.0853
  - Method **(ii)**: 0.1005
  - Method **(iii)**: 0.0731
  - Method **(iv)**: 0.0731
  - Method **(v)**: 0.0726
- **Conclusion:** The contrasts (adjusted using the Holm method) confirm that Method **(ii)** is significantly worse than all others ($p < 0.0001$). Method **(i)** is significantly worse than **(iii)**, **(iv)**, and **(v)** ($p < 0.0001$). Under this parametric testing framework, Methods **(iii)**, **(iv)**, and **(v)** are virtually tied and are not significantly different from one another ($p = 1.0000$ for all comparisons between them).

Tabulated Summary of Model Performance

Based on the statistical outputs, here is how the estimation methods handled the simulated observation error (sigma = 0.5) and overdispersion (od_mult = 0.5, theta_true = 10, theta_CV = 0.2).

Performance Rank	Estimation Method	Mean RMSE	Impact Summary
1 (Best)	Method **(v)**	0.0726	Highly robust against the simulated error regime. Statistically tied with **(iii)** and **(iv)** in the LMM , but non-parametric Wilcoxon tests suggest it has a slight edge.
2 (Tie)	Method **(iii)**	0.0731	Performs very well; statistically indistinguishable from **(iv)** and **(v)** in the mixed model.
2 (Tie)	Method **(iv)**	0.0731	Matches the performance of **(iii)** exactly in the estimated marginal means.
4	Method **(i)**	0.0853	Struggles moderately with the introduced overdispersion and error; significantly outperformed by **(iii)**, **(iv)**, and **(v)** in the LMM framework.
5 (Worst)	Method **(ii)**	0.1005	Most sensitive to the applied error parameters. Yielded the highest RMSE and performed significantly worse than all other methods across all tests.